Spatiotemporal Processing for Remote Sensing of Trapped Victims Using 4-D Imaging Radar

Zhi Li, Tian Jin, *Member, IEEE*, Lianlin Li, *Senior Member, IEEE*, Yongpeng Dai, Yongping Song, Yongkun Song, and Xiaolong Zhou, *Student Member, IEEE*

***Abstract*— It is of great importance to remotely sense trapped victims with radio signals in modern search and rescue after natural disasters such as earthquakes, avalanches, building collapses, and so on. Various radio sensors have been developed to date; however, they are hardly deployed to recognize efficiently the vital sign in long-distance, deep-coverage, and multisubject situations because the backscattered victim-critical radio signals are really weak and are nearly drowned in the ambient nonstationary noise and clutter. To tackle the formidable difficulty, we present a 4-D wideband microwave radar operating at 1.7–2.7 GHz and develop a spatiotemporal processing algorithm to fully explore the vital knowledge of victims in the 3-D spatial and 1-D temporal information. We conducted comprehensive field experiments in real postdisaster environments and demonstrated experimentally that our radio sensor can continuously monitor multiple survivors trapped under mounds of debris in real urban environments. Moreover, we demonstrate that the presented method can achieve a signal-to-noise-and-clutter ratio (SNCR) improvement of more than 20 dB even in the case of deep burial, which enables localizing the victims trapped in the order of 10 m and recognizing the survivors' vital states. We expect that the presented strategy may open an avenue for future remote life-rescuing and beyond in practical applications.**

***Index Terms*— Four-dimensional (4-D) imaging radar, search and rescue, spatiotemporal processing, vital signs sensing.**

I. Introduction

SENSING living people under the debris is an urgent technology that can be used in search-and-rescue tasks after avalanches, building collapses, earthquakes, and other disasters [1], [2], [3]. In the past 21 years (2001–2021), the global number of earthquake disasters accounted for 8% of the total number of disasters, with 761 582 deaths and 62% of disaster deaths [4], [5]. 90%–95% of the casualties are caused by the destruction and collapse of buildings. Searching and rescuing in time can help minimize the casualties. Many survivors are trapped alive after an earthquake or a building collapse but are difficult to be found under the debris. Rescue attempts are most effective in the first 72 h called golden 72 h, so detecting and localizing the victims have to be done as quickly as possible [6].

Microwave radars, seismic detectors, acoustic devices, and dogs have been exploited for searching trapped victims not visible by eyes or visual cameras. The employment of dogs has limitations to distinguish the living from the dead. Seismic detectors and acoustic devices require a quiet environment. By contrast, remote sensing by microwave radars is a promising technology in these applications. Electromagnetic waves transmitted by the radar, especially radio signals in low-frequency bands of several hundred MHz to several GHz, penetrate through walls, debris, and other obstacles to reach the targets. They are then scattered back to the radar's receiving antennas. The radar detects and localizes the human body by processing micromotion signals induced by cardiorespiratory activities [7], [8], [9], [10]. Up to now, most studies for vital signs detection were carried out in indoor or laboratory environments, with few measurements in real or simulated postdisaster environments. Our research is based on actual measured data from real environments in China's National Training Base for Urban Search and Rescue (CNTBUSR), which helps to verify the real effect of detecting vital signs in postdisaster environments and helps rescuers to develop detailed rescue plans to reduce secondary injuries.

Radar-based vital signs of the person under the debris are weak and nonstationary. They are associated with weak micromotion caused by respiratory and cardiac movements. Vital signs modulated in the radar echoes are submerged in noise and clutter due to the attenuation and multipath interference of the debris and walls, which makes it more difficult to detect the trapped victims. Therefore, detecting weak vital signs signals from noise and clutter is a key problem in postdisaster search and rescue using radars.

Various methods of radar-based vital signs detection through obstacles have been studied. Table I lists the review of remotely sensing trapped victims by radars published in academic papers. Nezirovic et al. [2] proposed a detection method based on discrete Fourier transform (DFT) and singular value decomposition (SVD) to detect the periodic motion caused

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Zhi Li is with the Department of Intelligent Sensing, National University of Defense Technology, Changsha 410000, China, and also with the Department of Communication, Army Engineering University of PLA, Nanjing 210007, China (e-mail: lizhi@nudt.edu.cn).

Tian Jin, Yongpeng Dai, Yongping Song, Yongkun Song, and Xiaolong Zhou are with the Department of Intelligent Sensing, National University of Defense Technology, Changsha 410000, China (e-mail: tianjin@nudt.edu.cn; daiyp3360@126.com; sypopqjkl@163.com; yongkunsong@nudt.edu.cn; zhouxiaolong@nudt.edu.cn).

Lianlin Li is with the State Key Laboratory of Advanced Optical Communication Systems and Networks, School of Electronics, Peking University, Beijing 100871, China (e-mail: lianlin.li@pku.edu.cn).

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by the breathing of the victim for low signal-to-noise-and-clutter ratio (SNCR) conditions. They separated the respiratory motion response of the trapped victim from nonstationary clutter at a building material dump. Xu et al. [11] proposed a life detection method based on fast Fourier transform (FFT) and constant false alarm ratio (CFAR) to improve the SNCR. Liu and Liu [12] employed the single-input single-output (SISO) stepped-frequency continuous wave (SFCW) radar to detect the respiratory rate (RR) of a human subject underneath layers of stacked clay bricks in the soil physics laboratory. Lv et al. [13] utilized the range interrelation to distinguish clutter from the respiratory motion through simulated earthquake debris. Yang et al. [14] proposed a method based on spectral kurtosis analysis to detect trapped victims behind a three-layer wall of 1 m using an impulse radio ultrawideband (IR-UWB) radar. Wu et al. [15] carried out a real experiment implementing an FFT-based method in low SNCR conditions such as through-wall and other similar conditions. Tayaranian Hosseini et al. [16] extracted parameters of the respiration behind two concrete piles using the time-varying representation of the Fourier coefficients. Yan et al. [17] implemented the detection of quasistatic trapped human beings by a monostatic UWB radar.

In the above research and other related work [14], [17], [18], [19], researchers mainly applied SISO radars to detect vital signs for their simple structures. However, the SISO radars can only detect the target and measure its range, and they cannot effectively remove the clutter in real scenarios surrounded by debris. To further localize the human subjects, multichannel technology can be exploited. Harikesh et al. [20] proposed a respiratory frequency detection method based on a multichannel Doppler radar, which can penetrate wood and glass to detect the angular positions and respiratory signals of multiple targets. Liang et al. [21] exploited a frequency-modulated continuous-wave (FMCW) multiple-input multiple-output (MIMO) radar combining with maximum spectrum peak, higher-order cumulant (HOC), coherence factor (CF), and attention mechanism to enhance through-wall vital signs. Distributed radar sensors were also exploited to detect and localize trapped victims in test and simulated environments with multiple reflections. Shargorodskyy and Herschel [22] proposed a radar concept of spatially distributed synchronized FMCW radar sensors to localize trapped victims and verified the algorithms in test and simulated environments with multiple reflections. Xia et al. [23] presented a multiple-observation-points-based vital sign searching method for trapped persons through the triangle composition of the measured ranges. The limitation is that it is not conducive to the coherent processing of multiple channels and is not easy to deploy in practice.

We address this challenge using a 4-D microwave radar, which can localize and monitor the vital signs of multiple subjects through obstacles. Because postdisaster rescue scenarios are much more complex than through a brick wall [24], reliable detection algorithms operated in a lower SNCR are required. Intuitively, the vital signs modulated in radio signal echoes distribute in space and change over time, thus sensing trapped victims is a dynamic process evolving in both space and time dimensions. Traditional radar image in the space domain offers an incomplete representation of the states of the survivors. It lacks an understanding of the temporal evolution of living bodies. However, 4-D microwave radar imaging can obtain image sequences as a 4-D continuum in 3-D spatial and 1-D temporal dimensions. It inspires us to make full use of the 3-D spatial and 1-D temporal information to improve the sensing performance. Furthermore, we carried out field experiments at CNTBUSR to validate the availability of effectively detecting and localizing the survivors trapped in postearthquake environments. The main contributions are summarized as follows.

1) A spatiotemporal remote-sensing approach based on 4-D radar imaging is proposed to detect trapped victims. The spatiotemporal joint processing takes advantage of the 3-D spatial and 1-D temporal information to improve the sensing performance.
2) The CF is used to suppress multipath interference in the 3-D spatial dimension. For sensing through debris and walls, the multipath overwhelms the echoes but has different distribution among various channels, which provides possibilities for suppressing the multipath clutter. As an adaptive spatial filter, CF weighting has a robust antimultipath ability in complex environments, so it can enhance the image of the target under debris.
3) The energy–entropy ratio (EER) is proposed to enhance vital body image in the temporal dimension. The quality of the received signal through debris or walls is relatively low, so it is difficult to detect the vital signs of the trapped persons. However, the information entropy difference between the vital signs signal and the noise signal is helpful to distinguish them. The proposed EER combining spectral entropy and subband energy can effectively suppress the interference of noise and clutter under the condition of low SNCR.

We organized the remainder of this article as follows. Section II makes a detailed description of the proposed method, including implementations of the CF processing in the spatial dimension and the EER processing in the temporal dimension. Then in Section III, the results of different scenarios for real environments at CNTBUSR are shown and evaluated. In Section IV, we discuss the results and the proposed method. In Section VI, we conclude this research.

TABLE I
RESEARCH REVIEW OF REMOTE SENSING TRAPPED VICTIMS WITH A MICROWAVE RADAR

Configuration	Frequency (GHz)	Subjects	Range (m)	Localization	Vital signs
SISO [2]	0.2-2	1	1.2	-	RR
SISO [11]	0.3-0.8	1	3.8	-	RR
SISO [12]	0.3-1.3	1	0.8-1.0	-	RR
distributed [22]	1.1-1.3	1	1.8	2D	-
multi-station [23]	0.3-1.7	1	3.6-3.7	3D	-
4D [This work]	1.7-2.7	2	2.2-6.3	3D	RR

II. TRAPPED VICTIMS DETECTION BY 4-D IMAGING RADARS

Radar imaging has the ability to obtain target images by transmitting and receiving electromagnetic waves at various

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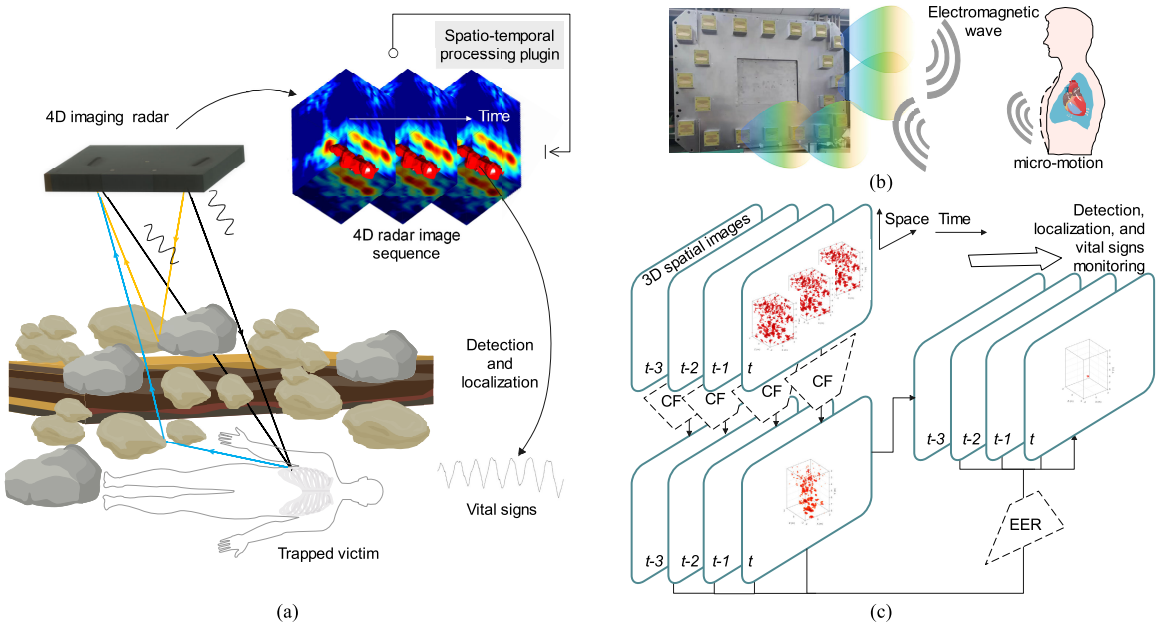


Fig. 1. Sketch of detecting and localizing trapped victims using the 4-D-imaging microwave radar. (a) Illustration of the working scene. The blue lines indicate the multipath, and the orange lines indicate the background clutter. (b) Principle of vital signs sensing exploiting the wideband MIMO array. Each transmitting element emits a wideband signal, and all the receiving elements receive the echoes modulated by the micromotion of a living person. (c) Pipeline of the spatial and temporal processing for the radar images.

frequencies and observation angles. When operating at low frequencies from several hundred MHz to several GHz, occluded targets can be seen in the image. Fig. 1(a) illustrates the schematic of localizing and sensing trapped victims by a 4-D microwave radar. In this research, we developed a 4-D-imaging microwave radar with an MIMO array transmitting SFCW radio signals as shown in Fig. 1(b). The obtained radar images, especially 4-D image sequences including spatiotemporal information, are conducive to detecting and localizing trapped victims under debris or behind walls. The pipeline of the spatial and temporal processing for the 4-D radar images is depicted as Fig. 1(c). First, the radar imaging algorithm is employed to echo data of multiple channels to form 4-D sequential radar images. Then the spatiotemporal processing is performed in the 3-D space and 1-D time. For every frame in the sequence, the image is weighted by CF to suppress multipath clutter. Then the EER is calculated along the temporal dimension for each voxel of the image sequence. Thus, weak vital signs are enhanced to reduce environmental noise and clutter. Finally, the trapped victims are detected by applying detection and localization algorithms.

A. Spatiotemporal Sequential Data Acquisition by 4-D Radar Imaging

Radar imaging provides a quantitative reconstruction of the dynamic targets concerning the location and variation estimation. The imaging can be performed as an electromagnetic inverse scattering problem [25]. The electric field $E(\boldsymbol{r})$ at spatial position $\boldsymbol{r}$ in the target area satisfies the Helmholtz equation

$$\nabla^2 E(\boldsymbol{r}) + k^2 E(\boldsymbol{r}) = -j\omega\mu J(\boldsymbol{r}) \tag{1}$$

where $k = \omega(\mu\varepsilon)^{1/2}$ represents the wavenumber, ω denotes the angular frequency, μ and ε represent the permeability and permittivity of the background medium, respectively, and $J(\boldsymbol{r})$ denotes the current density. Under the Born approximation, we can assume that the internal field of the target is approximately equal to the incident field [26], so the relationship between the scattered field and the contrast function can be written by the linear integral equation as follows:

$$E_s(\boldsymbol{r}', k) = k^2 \iiint_{\mathcal{D}} G(\boldsymbol{r}, \boldsymbol{r}', k)\, E_{\text{inc}}(\boldsymbol{r}, k)\, \chi(\boldsymbol{r})\, \mathrm{d}\boldsymbol{r} \tag{2}$$

where $\chi(\boldsymbol{r}) = \varepsilon_t(\boldsymbol{r})/\varepsilon - 1$ denotes the contrast function related to the reflectivity of the targets in the investigation region $\mathcal{D}$. It accounts for the difference between the targets' permittivity $\varepsilon_t(\boldsymbol{r})$ and the background medium. $E_{\text{inc}}(\boldsymbol{r}, k)$ is the incident field at position $\boldsymbol{r}$. $G(\boldsymbol{r}, \boldsymbol{r}', k)$ represents Green's function accounting for the field at position $\boldsymbol{r}'$ radiated by the equivalent source at position $\boldsymbol{r}$. The discretized matrix form of the above propagation model can be expressed as follows:

$$\boldsymbol{E}_{\text{s}} = \boldsymbol{H}\boldsymbol{\chi} + \boldsymbol{\nu} \tag{3}$$

where the scene is divided into N voxels and $\boldsymbol{E}_{\text{s}}$ is the sampled data of the scattering field $E_{\text{s}}(\boldsymbol{r}', k)$. Green's function and the incident field are key ingredients to construct the kernel of the sensing matrix $\boldsymbol{H}$. By normalizing the measured field, $\boldsymbol{H}$ merges the incident field with Green's function $G(\boldsymbol{r}, \boldsymbol{r}', k)$ in (2). Considering the propagation model, $\boldsymbol{H}$ is a linear operator with entry given by $H_{mn} = E_m^T(\boldsymbol{r}_n) \cdot E_m^R(\boldsymbol{r}_n)$, where $E_m^T(\boldsymbol{r}_n)$ and $E_m^R(\boldsymbol{r}_n)$ are the fields correspond to Green's functions for the mth transmitter/receiver pair and the nth voxel located at $\boldsymbol{r}_n$. Fig. 1(b) shows the M_T transmitters and M_R receivers detecting the micromotions of a person by synchronously transmitting and receiving backscattered radio signals with L

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samples. The backscattered signals are then arranged into an $M \times 1$ vector $\boldsymbol{E}_s$, where $M = M_T M_R L$ denotes the number of measurements. $\boldsymbol{\nu} \in \mathbb{C}^M$ denotes the measured noise and clutter.

The imaging process aims to estimate the reflectivity map of the targets. There are numerous methods to obtain the image by solving the forward process. As $\boldsymbol{H}$ usually has a large size and is rarely an invertible matrix in real applications, it is computationally intractable to obtain the optimal estimation in the sense of least square

$$\hat{\boldsymbol{\chi}} = \left(\boldsymbol{H}^\top \boldsymbol{H}\right)^{-1} \boldsymbol{H}^\top \boldsymbol{E}_{\mathrm{s}} \tag{4}$$

where $^\top$ denotes the conjugate transpose. Instead, an accurate, simple, and robust solution for the imaging model is the back-projection (BP) imaging algorithm [27]. The algorithm behaves as a matched filter with the solution $\hat{\boldsymbol{\chi}} = \boldsymbol{H}^\top \boldsymbol{E}_{\mathrm{s}}$. This algorithm is robust for complex scenarios and simple for subsequent spatiotemporal processing. It can also be implemented with FFT and parallel computing to speed up the calculation. For real-time measurements with multiple transmitting and receiving elements, the estimated $\hat{\boldsymbol{\chi}}$ in the reconstructed image at time t can be expressed as follows:

$$I(\boldsymbol{r}, t) = \sum_{m_T=1}^{M_T} \sum_{m_R=1}^{M_R} \sum_{l=1}^{L} S_{m_T m_R}(f_l, t) e^{jk_l \left[R_{m_T}(\boldsymbol{r},t) + R_{m_R}(\boldsymbol{r},t)\right]} \tag{5}$$

where $\boldsymbol{r} = (x, y, z)$ represents the discrete 3-D spatial coordinates, and $t = p\Delta T$ represents the discrete time at the pth instant with the sampling interval ΔT. $S_{m_T m_R}(f_l, t)$ is the radar echo from the m_Tth transmitting element to the m_Rth receiving element with the lth frequency f_l. L is specified as the number of frequency points for the echo data sampled in the frequency domain. $k_l = 2\pi f_l / V$ is the corresponding wavenumber, where V represents the propagation velocity of the electromagnetic wave. $R_{m_T}(\boldsymbol{r}, t)$ and $R_{m_R}(\boldsymbol{r}, t)$ represent the distance between the imaging voxel and the array elements.

After applying the imaging algorithm to the raw radar data, we get a sequence of 4-D spatiotemporal images in 3-D space and 1-D time. The image sequence in (5) contains the micromotion displacements. According to [24], the voxel at the chest wall is further expressed as follows:

$$I(\boldsymbol{r}, t) = L M_T M_R \chi(\boldsymbol{r})\, e^{-j2k_c \rho(\boldsymbol{r}, t)} \tag{6}$$

where k_c is the wavenumber corresponding to the central frequency. $\rho(\boldsymbol{r}, t)$ is the body displacement in the direction in which the radar points to the target located at $\boldsymbol{r}$.

The 4-D radar image sequence is illustrated in Fig. 1. It contains the spatial information in the individual 3-D image, and the time-series information corresponding to the vital signs can be extracted by the sequential processing of the images. In the application of detecting trapped victims, the multipath interference overwhelms the radar echoes, and the vital signs are too weak to detect directly.

Considering the noise $\nu(\boldsymbol{r}, t)$ and the multipath interference, we can present the image sequence as follows:

$$I(\boldsymbol{r}, t) = L M_T M_R \chi(\boldsymbol{r})\, e^{-j2k_c \rho(\boldsymbol{r}, t)} + \nu(\boldsymbol{r}, t) + \sum_{\gamma} \sum_{m=1}^{M_T} \sum_{n=1}^{M_R} \sum_{l=1}^{L} b_\gamma \chi(\boldsymbol{r})\, e^{-jk_l R_{mn\gamma}(\boldsymbol{r}, t)} \tag{7}$$

where $R_{mn\gamma}(\boldsymbol{r}, t)$ represents the γth multipath related to the human body at $\boldsymbol{r}$, and b_γ is the corresponding multipath contribution coefficient. The first term in (7) is the useful contribution, while the last one is the clutter term. Low-frequency electromagnetic waves can propagate through materials and objects with low attenuation, but the electromagnetic propagation speed reduces due to the medium. The propagation can be modeled through an equivalent propagation path that replaces the actual path with an equivalent factor such as equivalent permittivity and wavenumber [26], [28]. It regards the propagation in the scenario as occurring in a medium with an equivalent and spatially varying permittivity. The wavenumber k is substituted by the equivalent wavenumber in the expressions. Thus, the direct path echoes can be focused on the target positions by proper compensation. Conversely, the multipath disperses for different transceivers, frequencies, and scatters at different moments, thus consequently the related echo components are erroneously accumulated. Fig. 2(a) shows the direct propagation path and multipath at an instant and different instants. It inspires us to take advantage of the spatiotemporal differences in the image sequence to sense the targets of interest.

B. CF in Spatial Dimension

The concept of CF has been used for sidelobe artifacts mitigation [29] and ghost suppression in radar imaging applications [30], [31]. It takes advantage of the spatial differences of radar echoes from different transceivers in the aperture. Inspired by the coherent processing, we come up with the idea of using CF to enhance the radar images by suppressing the multipath artifacts. In complex environments such as through debris or wall, the multipath influence has a greater impact on the radar image than ghosts and sidelobes. The irregular multipath results in interference dispersed around the target. Since the 4-D radar has numerous channels, the multipath interference of each channel is different, so the interference can be reduced while the target remains.

The CF enhancement method utilizes differences in multichannel echoes to calculate the weighting coefficients. Many formulas have been proposed to calculate the weighting coefficients, such as phase coherence and sign coherence [31], [32]. The classical formula of CF can be expressed as follows:

$$\mathrm{CF}(\boldsymbol{r}, t) = \frac{\left| \sum_{m_T=1}^{M_T} \sum_{m_R=1}^{M_R} I_{m_T m_R}(\boldsymbol{r}, t) \right|^2}{M_T M_R \sum_{m_T=1}^{M_T} \sum_{m_R=1}^{M_R} \left| I_{m_T m_R}(\boldsymbol{r}, t) \right|^2} \tag{8}$$

where $I_{m_T m_R}(\boldsymbol{r}, t)$ represents a 4-D spatiotemporal sequential subimage obtained from the m_Tth transmitting and the m_Rth receiving channel. As the BP algorithm shows in (5), the subimage can be calculated in the intermediate process as

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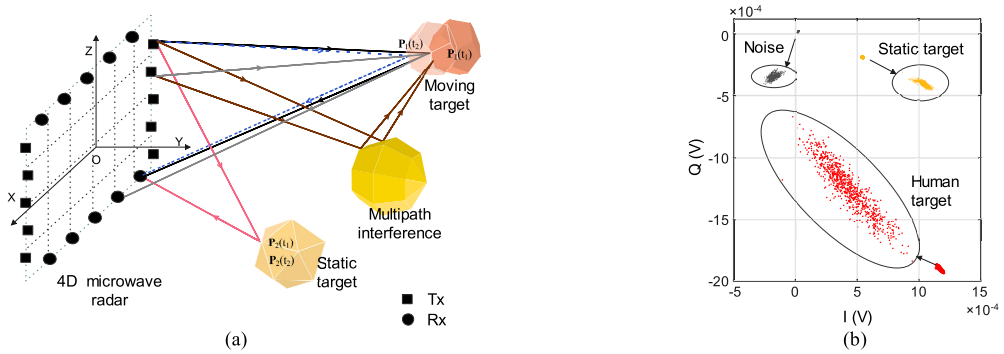


Fig. 2. Sensing principle of 4-D-imaging microwave radar with an MIMO array. (a) Spatial and temporal differences of distinguishable targets. The direct propagation paths (solid black and gray lines) vary with different Tx/Rx pairs regularly, while the multipath interference (solid brown lines) fades irregularly. The propagation path of a moving target (dashed blue line) changes over time, while it keeps stationary for a static target (solid red line). (b) Constellation of the demodulated signals extracted from the human target, static target, and noise. The scatter points are zoomed in as indicated by the arrows.

$I_{m_T m_R}(\boldsymbol{r},t) = \sum_{l=1}^{L} S_{m_T m_R}(f_l,t)e^{jk_l[R_{m_T}(\boldsymbol{r},t)+R_{m_R}(\boldsymbol{r},t)]}$, which implies the complex-valued contribution of each channel to the voxel at $\boldsymbol{r}$. Moreover, other kinds of calculating the subimage such as a combination of different channels or frequencies [31] are also available.

In contrast to sidelobe and ghost suppressions, the exploited CF aims to reduce the multipath artifacts in the whole image. It helps to reveal human targets submerged in the overwhelmed multipath interference induced by the complex wall/debris environment. Within the target area, the subimages of multiple channels are similar, so the CF value is larger. For the multipath clutter, the imaging results of different channels are different, which leads to a smaller CF value. Therefore, an adaptive 3-D spatial filter can be formed to suppress the interference of multipath clutter. The CF-enhanced image is obtained by pointwise multiplying the weighting coefficients at each spatiotemporal point

$$I_{\mathrm{CF}}(\boldsymbol{r},t) = I(\boldsymbol{r},t)\,\mathrm{CF}(\boldsymbol{r},t). \tag{9}$$

Other CF-based methods, such as phase CF (PCF) [31], can also be applied by replacing (8) with the specific definition. In this article, we use the basic CF for enhancement.

Consequently, change detection can be used to detect vital signs caused by breathing and cardiac motion. However, the vital signs are very weak in the overwhelming noise. It is difficult to effectively distinguish targets from noise. Therefore, it is necessary to design an appropriate algorithm for weak vital signs detection.

C. EER in Temporal Dimension

The micromotions due to cardiopulmonary activities of the trapped person lead to different features in the image compared with static targets and noise over time evolution. Fig. 2(a) also gives the propagation paths of a moving target $\boldsymbol{P}_1$ and a static one $\boldsymbol{P}_2$. Likewise, the propagation path of the moving target is different from that of the static target at different samplings time (i.e., t_1 versus t_2).

The extracted complex signals due to a radar image of a trapped human under the debris, a static stone, and the noise of the interspace are plotted in Fig. 2(b). As the micromotions of the trapped victims are usually sensed under thick obstacles, the extracted vital signs are blurred and difficult to be distinguished. However, the distributions of the complex signals are slightly different in shape and size. It is because the temporal evolution of the points related to vital signs has regular fluctuations while the noise is random. With the potential of distinguishing information from noise, entropy has been applied to describe the regularity of the signal [33]. The more regular the signal is, the smaller the corresponding entropy is, and vice versa. It means that the entropy could characterize the periodicity and cyclostationarity of a signal [34].

For each spatial point in the radar image sequence, the related time-series signal has different information entropy between human subjects and environmental noise. Usually, human activity makes the intrinsic frequency band implied in the radio signal diverse from the noise. For the micromotions caused by respiration and heartbeat, the spectrum is mainly distributed between 0.1 and 3 Hz, while the noise is dispersed much more widely. Therefore, it is intuitive to distinguish vital signs from noise in the frequency domain. Especially, spectral entropy is proper because it indicates the disordered state of the spectrum or frequencies corresponding to the signal or system. It has been used in the fields of gear fault detection, speech detection, and wireless communication signal processing [33], [35]. A method based on spectral entropy has also been successfully used in spectrum sensing in cognitive communication to detect weak signals under a low SNR [35]. When the frequency components of the signal are simple, the power spectrum is concentrated in the frequency domain, which results in a smaller entropy. If the signal is chaotic, the power disperses over the spectrum, which results in a larger entropy. As a result, spectral entropy has been a quantitative indicator of the complexity of the signal in the frequency domain.

To enhance the living body and eliminate the residual artifacts, combining signal energy with information entropy can further improve the detection performance under low SNR conditions. Especially in the application of penetrating debris, there are many stationary objects in the detection area. The stationary background clutter overwhelms the human target in the reflectivity of the radar image. However, its vibration energy is much lower than the human target due to physiological micromotion activities. Therefore, the EER

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weighting algorithm is proposed for distinguishing the vital body from the environmental noise.

The proposed EER combines spectrum entropy and signal energy to further improve the robustness. The EER can be calculated by the steps as follows.

1) For each voxel $\boldsymbol{r}$, the time–frequency power spectrogram corresponding to $I(\boldsymbol{r},t)$ is represented as $X(\boldsymbol{r},t,f) = |S(\boldsymbol{r},t,f)|^2$, where $S(\boldsymbol{r},t,f)$ is the time–frequency representation of $I(\boldsymbol{r},t)$ calculated by the short-time Fourier transform (STFT) or other time–frequency representation.
2) To compute the instantaneous spectral entropy given a time–frequency power spectrogram $X(\boldsymbol{r},t,f)$, the probability distribution at time t is
$$P\left(\boldsymbol{r},t,f_i\right) = \frac{X\left(\boldsymbol{r},t,f_i\right)}{\sum_{f_i\in\mathcal{F}} X\left(\boldsymbol{r},t,f_i\right)} \tag{10}$$
where f_i represents the ith discrete frequency in the frequency band $\mathcal{F}$.
3) The spectral entropy of the 4-D radar image sequence at time t is calculated by using a standard formula of Shannon entropy
$$H\left(\boldsymbol{r},t\right) = -\sum_{f_i\in\mathcal{F}} P\left(\boldsymbol{r},t,f_i\right)\log P\left(\boldsymbol{r},t,f_i\right). \tag{11}$$
4) The EER is defined as $E_h(\boldsymbol{r},t) = (E(\boldsymbol{r},t))/(H(\boldsymbol{r},t))$, where $E(\boldsymbol{r},t) = \sum_{f_i\in\mathcal{F}_v} X(\boldsymbol{r},t,f_i)$ is the energy of the specified spectrum at time t, and $\mathcal{F}_v$ indicates the specified frequency range of the micromotion signal which is mainly distributed between 0.1 and 3 Hz for human beings.
5) Then the image is filtered by pointwise multiplication of the EER
$$I_{\text{EER}}\left(\boldsymbol{r},t\right) = I_{\text{CF}}\left(\boldsymbol{r},t\right)E_h\left(\boldsymbol{r},t\right). \tag{12}$$

By applying the 4-D sequential CF and EER, the images are significantly enhanced. After this plugin, subsequent radar image processing approaches such as detection and localization methods can be applied to the enhanced images for various purposes.

III. Earthquake Field Experiments

We carried out field experiments at CNTBUSR to verify the effectiveness of using 4-D-imaging microwave radar for sensing and searching trapped victims under debris. The training base is located at the foot of Fenghuangling Mountain in northwest Beijing, China, and the latitude and longitude coordinates are (116.104 092° E, 40.115 964° N). There are almost all the situations of earthquake-affected training buildings and can drill almost all search and rescue scenarios. The experimental environment is shown in Fig. 3. The collapsed building has three floors with a height of 3 m each. The collapsed part forms a triangular spatial structure called the Triangle of Life, and on the west side of the building is the construction debris.

We conducted experiments in three scenarios. Scenario 1 is in the interior of the partially collapsed building. The radar is placed on the third floor, and the trapped victim lies calmly on the first floor. The floors are made of reinforced concrete slabs with a thickness of 15 cm. The purpose of this experiment is to verify the ability to detect trapped victims from a long distance. Scenario 2 is under the postdisaster debris of the collapsed building. The victim is trapped about 2.2 m below the debris of the broken concrete slabs. We design this experiment to verify the ability to penetrate thick debris in a complex environment. Scenario 3 shows the situation with a triangle of life. The Triangle of Life is a space formed by the collapse of roofs and walls of buildings. The use of this space can help to avoid injuries and save lives in an earthquake or similar disasters. In this scenario, two persons situate about 1 m behind two parallel walls. The two walls are each 12-cm-thick with an interval of 0.8 m. This experiment aims to verify the ability to detect multiple subjects.

TABLE II
Key Parameters of the 4-D Imaging Radar Prototype

Parameters	Value
transmitting wave mode	SFCW
central frequency	2.2 GHz
bandwidth	1 GHz
frequency step	4 MHz
power of transmitted signal	20 dBm
frame rate	20 Hz
antenna element number	10 Tx, 10 Rx
array size	0.8 m × 0.6 m
antenna element gain	5 dB

The exploited 4-D imaging radar prototype for detecting trapped victims is also shown in Fig. 3. It is a portable all-solid-state radar. The block diagram in Fig. 4 shows the system structure of the radar. The radar transmits low-power radio frequency signals of 100 mW, which is less than a cellphone. The central frequency of the transmitted SFCW is 2.2 GHz with a bandwidth of 1 GHz, which can penetrate through building materials such as bricks and reinforced concretes [36]. It uses a $10T \times 10R$ planar MIMO array with bowtie antennas to form a virtual array with 100 equivalent elements. It takes 50 ms for all channels to complete one scan, so it can work continuously in real time at a frame rate of 20 Hz. The main parameters are listed in Table II.

IV. Experimental Results Analysis

After recording the data, we applied the proposed method to get the imaging results. Fig. 5 shows one frame of the image sequence before and after the processing by CF and EER. We plot the 3-D images in Cartesian coordinates with the origin at the radar center and uniform mesh grids of 0.1 m. Each row corresponds to one scenario, and the leftmost column of each row depicts the sketch of the scenario.

For scenario 1, the image before processing is dominated by clutter due to the diffraction of the three floors. Moreover, the echoes of the person attenuate significantly for the long propagation distance, which results in weak intensities around the person in the image. The clutter and noise are reduced after the CF operation and the human subject appears around the region (0.2, 6.4, and 0.1 m) after the subsequent EER processing.

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Fig. 3. Experimental scenarios for detecting trapped victims at CNTBUSR. The satellite map (left) and the field photo (right) both show the environment. The three different scenarios are zoomed in the white frame of the field photo.

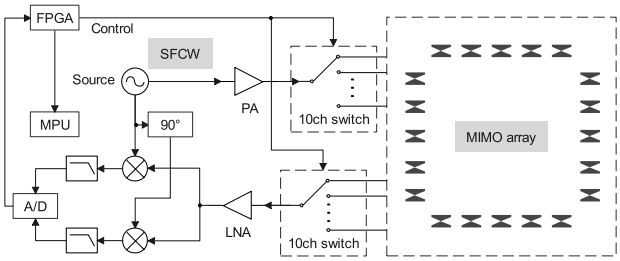


Fig. 4. System structure of the 4-D imaging radar. The ten transmitting elements emit radio signals in turn, and the ten receiving elements concurrently receive echoes scattered from the targets.

For scenario 2, the environment is more complex than in scenario 1, and the multipath interference severely affects the detection of the trapped human. However, the proposed method successfully reveals the subject around $(-0.1\,,\ 3.5\,,$ and $-\ 0.2\,\mathrm{m})$. Because of the dense debris medium, the position of the human body in the image is deeper than the actual position. Nevertheless, it does not affect the discovery and general localization of the trapped victim. Additionally, as the spatiotemporal processing is performed on each frame of the echoes, we can recover the videos using radar image sequences at a frame rate of $20\,\mathrm{Hz}$, which further highlights the ability of dynamic perception.

Scenario 3 is validated for the condition of sensing multiple subjects. The results in the third row show that two subjects appear at the regions around $(-0.6\,,\ 2.5\,,\ -\ 0.5\,\mathrm{m})$ and $(0.3\,,\ 2.2\,,\ -0.5\,\mathrm{m})$, respectively. The localization results agree with the actual positions of the two persons about $1.0\,\mathrm{m}$ apart. The localization results also show their relative position in 3-D space.

For penetrating detection through wall/floor/debris, multipath clutter seriously interferes with human detection. As shown in the second column of Fig. 5, the image of the trapped victim in scenarios 1 and 2 are severely blurred. There are too many false targets to distinguish the human targets. The spatial filtering based on CF effectively suppresses the multipath clutter and enhances the target. The experimental results in the third column show that this spatial processing is effective compared with the second column, especially in the case of through floor and debris imaging.

To better show how the EER works, we observe the time series for different spatial points of scenario 2 as shown in Fig. 6. For the imaging results of scenario 2, we focus on the detected voxel and a randomly selected voxel outside the region of the subject. Both the extracted micromotion signals are full of noise and interference as shown in Fig. 6(a) and (d). However, from the spectrum entropy in Fig. 6(c) and (f), the detected voxel has a lower entropy in the frequency range of 0–2 Hz indicating that it contains information in this band, while the randomly selected voxel mainly contains noise for high entropy in all the bands. Therefore, combining the energy of the signal, the EER in Fig. 6(b) is much larger than that of Fig. 6(e).

To further analyze the performance, we evaluate the proposed method from three aspects: SNCR improvement, reliability analysis, and vital signs extraction.

A. SNCR Improvement

The visual effect of the proposed method has been shown in Fig. 5. We further carried out quantitative analysis by calculating the SNCR. For the images at time t, the SNCR is defined as follows:

$$\mathrm{SNCR}\,(t) = 10\lg\frac{\sum_{\boldsymbol{r}\in\mathcal{X}_{\mathrm{tar}}}|I\,(\boldsymbol{r},t)|^2/N_{\mathrm{tar}}}{\sum_{\boldsymbol{r}\in\mathcal{X}_{\mathrm{noi}}}|I\,(\boldsymbol{r},t)|^2/N_{\mathrm{noi}}} \tag{13}$$

where $\mathcal{X}_{\mathrm{tar}}$ represents the target region with N_{tar} voxels, and $\mathcal{X}_{\mathrm{noi}}$ represents the noise region with N_{noi} voxels. The detection regions are discretized into cubic voxels with uniform grids of $0.1\,\mathrm{m}$, thus the total voxel number of each scenario is determined accordingly. N_{tar} is the number of voxels occupied by the region of the human body with a typical size of $2\times 0.6\times 0.4$ m, and N_{noi} refers to the voxel number of the region outside the human body. For a duration of K frames, the average is defined as $\overline{\mathrm{SNCR}} = (1/K)\sum_t \mathrm{SNCR}(t)$.

For the three different scenarios, Fig. 7 shows the average SNCR of the original 3-D image and the processed results.

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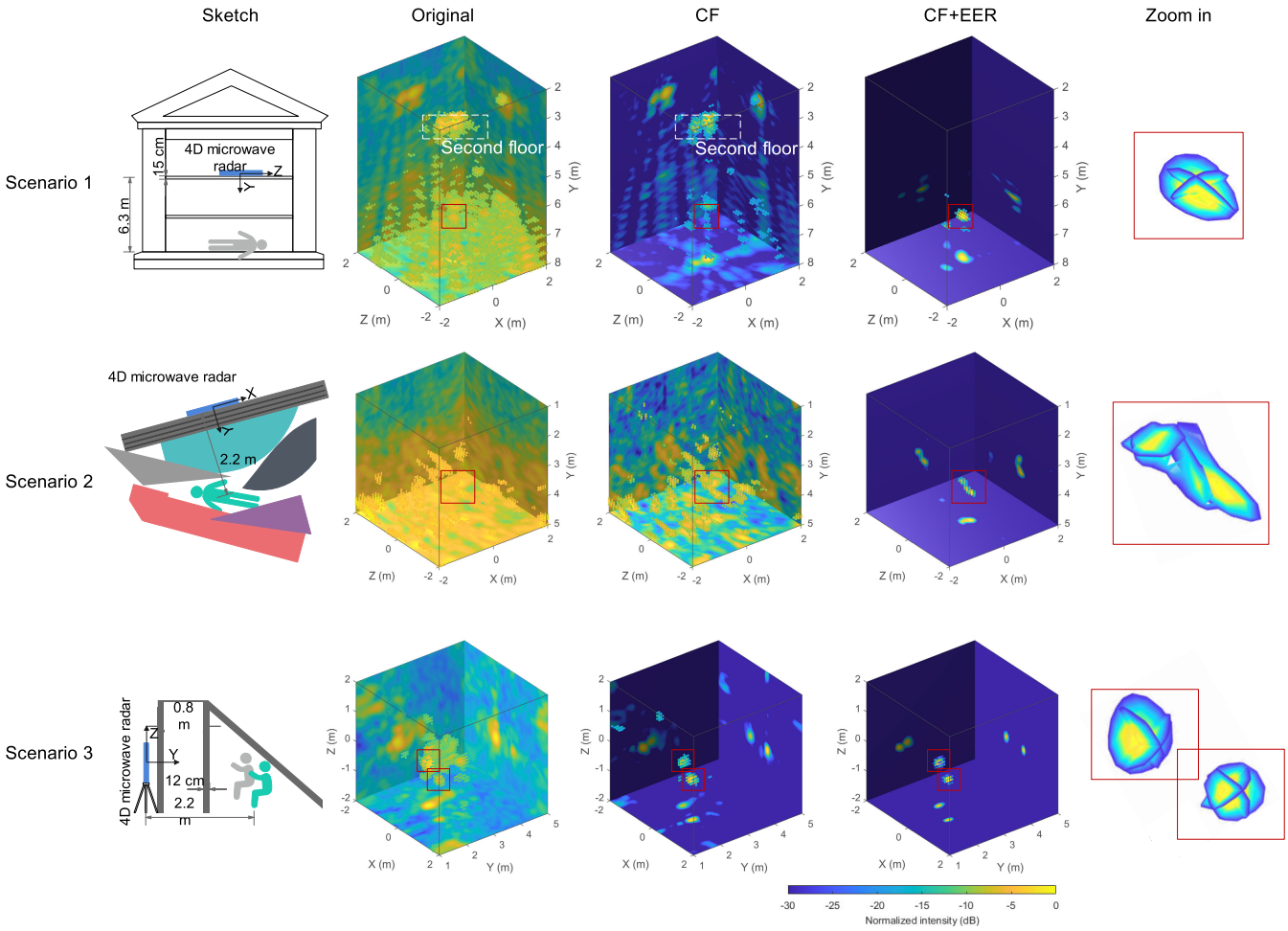


Fig. 5. Imaging results of the three scenarios. The second column shows one frame of the original 4-D radar image sequence at an instant for each scenario. The third column shows the corresponding frames after CF processing. The fourth column shows the frames after CF and EER processing. 3-D point clouds are shown with the scattering intensity. XY, YZ, and ZX projections of the 3-D images are plotted on the three orthogonal planes. The last column zooms in on the areas related to the trapped persons as 3-D slices.

CF and EER both improve the SNCR separately, and the spatiotemporal joint processing further improves the SNCR. Fig. 7 also shows that the improvements are different with CF or EER processing for the three scenarios. The reason is that the multipath effects are diverse in different environments, so the improvements by CF vary considerably. In scenario 2, the concrete debris fills the imaging area, which leads to a more complex multipath effect than in the other two scenarios. However, it still improves about $10\,\mathrm{dB}$ compared with scenario 1 for $20\,\mathrm{dB}$ and scenario 3 for $30\,\mathrm{dB}$. On the other hand, the improvements by EER are all about $20\,\mathrm{dB}$ for the three scenarios. The reason is that EER operates on the time series of the vital signs in the temporal dimension which are similar. Ultimately, the spatiotemporal processing greatly improves the SNCR compared with the original images.

B. Reliability Analysis

To verify the reliability of the proposed method, we calculated the probability of detection (Pd) for various SNRs compared with different processing methods. Successful detection is defined as if the estimated positions are consistent with the simulated ones. The Pds by the proposed method compared with the FFT-based [11] and HOC-based [21], [24] algorithms are shown in Fig. 8. The algorithms are performed on the simulated echoes with additive white Gaussian noise. The vital sign of the target is simulated according to [10] with a respiratory amplitude of $10\,\mathrm{mm}$ and a heartbeat amplitude of $2\,\mathrm{mm}$. The related respiration rate and heart rate are 15 breaths per minute (brpm) and 70 beats per minute (bpm), respectively. The target locates at a distance of $2\,\mathrm{m}$ in front of the radar center, and the simulation parameters of the radar are the same as that of the 4-D-imaging microwave radar employed in the field experiments. The performance is evaluated utilizing Monte-Carlo simulation. The SNR is defined as the ratio of the vital sign component to the noise in the simulated radar echo signal corresponding to (3).

The results are shown in Fig. 8. All the algorithms perform well for SNRs above $-30\,\mathrm{dB}$. When the SNR is lower, the proposed method has the best detection performance compared with the other ones. We also give the results of comparison with different spatial processing (with or without CF) and temporal processing (FFT, HOC, and EER). Overall, EER

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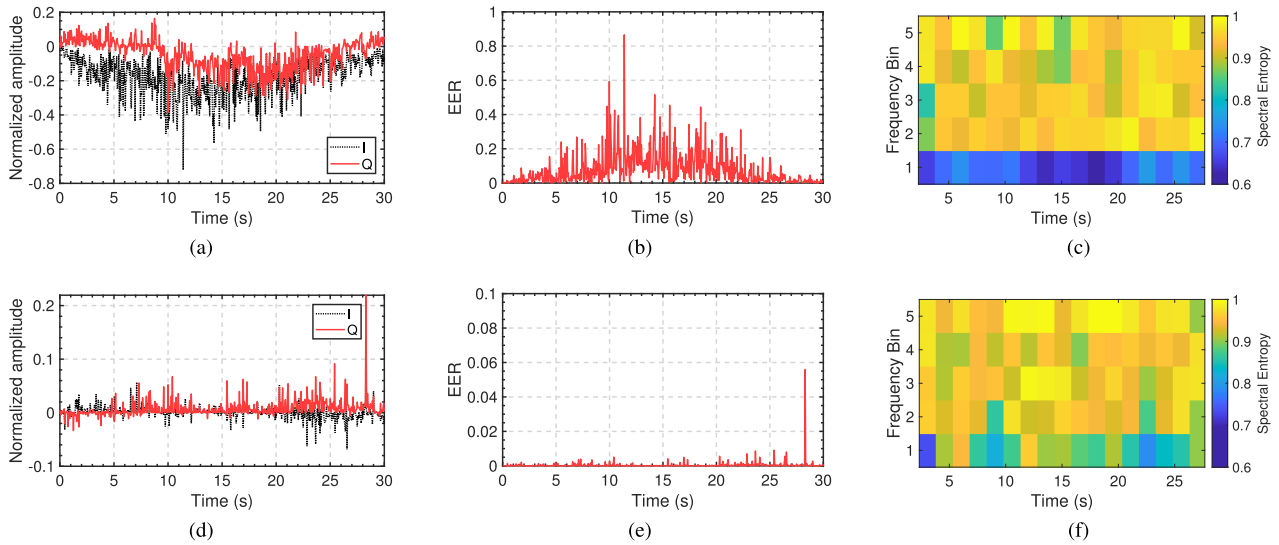


Fig. 6. Time series results of (a)–(c) detected subject voxel and (d)–(f) randomly selected background voxel in scenario 2. (a) and (d) Real (I) and imaginary (Q) components of the extracted signals. (b) and (e) Corresponding EERs normalized by the amplitude of the 4-D images. (c) and (f) Spectrum entropy for five frequency bins with equal intervals from 0 to 10 Hz.

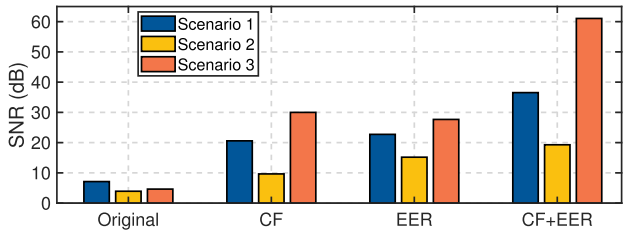


Fig. 7. Improvements of the SNCR for different scenarios with different spatiotemporal processing.

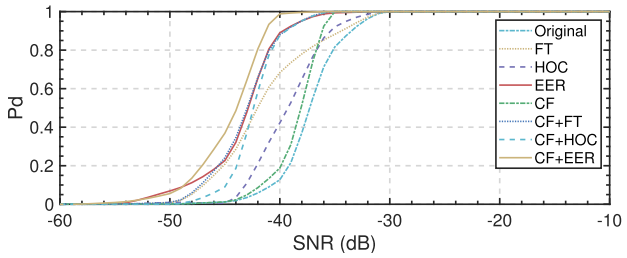


Fig. 8. Detection probability for various SNRs comparing with different processing methods.

combined with CF outperforms the FFT-based algorithms and the HOC-based algorithms.

C. Vital Signs Extraction

Besides the radar images, extracting the vital signs of the trapped persons is emergent in search and rescue. Remote vital signs measurement is based on micromotion detection using microwaves. The scattered radio signals from human bodies carry motion information of cardiopulmonary activities, including inhalations and exhalations of the lungs and contractions and relaxations of the heart. Typical displacement of the chest wall induced by breathing ranges from 4 to 12 mm. The surface movements of the thorax are measured in sequence, so the arrival distance of the transmitted radio signal varies accordingly. Therefore, we can measure the physiological signals by extracting slight variations in the radar image sequence.

The displacement induced by the cardiopulmonary activity is proportional to the phase of the image according to (6). Therefore, after detecting and localizing the victims, we can extract the body displacement through the sequential phase of the voxel as a time series [24]

$$\rho(t) = -\text{Phase}\left(I(\boldsymbol{r}, t)\right) \cdot \frac{1}{2k_c} \tag{14}$$

where Phase$(\cdot)$ represents the phase calculation function such as the arc-tangent function and the extended differentiate and cross multiply (DACM) algorithm [24]. Thus, the sequential unwrapped phase of the voxel detected at the specific position $\boldsymbol{r} = \boldsymbol{r}_0$ can be calculated along the time dimension. The displacement mixes the respiration and heartbeat signals, which can be further separated by signal-processing algorithms such as adaptive filters [10].

The vital signs waveforms are extracted as shown in Fig. 9 by observing the time series of the detected spatial locations. For scenario 1, the real and imaginary parts of the extracted vital signs indicate the respiration motion of the trapped person as shown in Fig. 9(a). The corresponding RR is $18.9\,\text{brpm}$.

For scenario 3, we further extract the waveforms of the vital signs for both persons by tracking the phases of the voxels. The results are shown in Fig. 9(b). Then we separate the respiratory pattern by an infinite impulse response (IIR) filter with a passband of 0.1–0.8 Hz. As shown in Fig. 9(b), the respiration curves of the two persons are different in amplitude and period. The respiratory pattern of person 1 (Respiration 1) has a larger amplitude and a shorter period, which indicates that person 1 takes deep breaths. In contrast, the respiratory pattern of person 2 (Respiration 2) has a smaller amplitude and a longer period, which indicates that person 2 takes shallow breaths. The related RRs are 16.6 and $22.3\,\text{brpm}$, respectively. These monitoring results are consistent with the actual situation.

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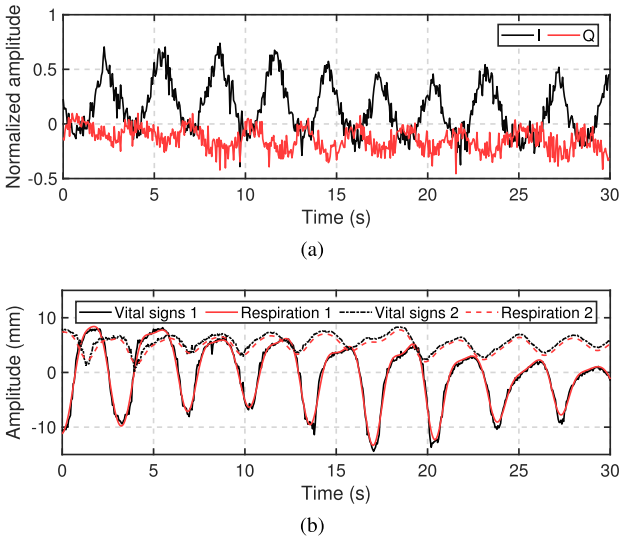


Fig. 9. Extracted vital signs waveforms in (a) scenario 1 and (b) scenario 3. The real (I) and imaginary (Q) parts of the extracted signal are plotted in (a). The vital signs in (b) are extracted by the phases of the complex signals. The respiration signals indicated as Respirations 1 and 2 show the different respiratory patterns of the two subjects.

For scenario 2 with an extremely low SNCR, the extracted vital signs are rather irregular, as shown in Fig. 6(a). Even though it is difficult to extract the vital signs under these conditions with heavy debris, we successfully detected and localized the victims, as shown in Fig. 5.

V. Discussion

This spatiotemporal processing method has low computational complexity, strong robustness, and good performance in different trapped scenarios. For the hardware we exploited in the experiments, the radio signal data are transferred to a laptop with an Intel Core i7 processor working at a frequency of 2.3 GHz and a memory of 16 GB. The CF-based spatial processing is real time together with the imaging process, and the EER-based temporal processing is quasi-real time with a delay of about $150\,\mathrm{s}$ for searching a region of $96\,\mathrm{m}^3$. Since the parallel processing structure is used for each voxel in the image sequence, the timeliness of processing can be further improved by graphic processing unit (GPU) acceleration.

In addition to the spectral entropy used in this article, we can also apply other entropy functions, such as sample entropy, approximate entropy, fuzzy entropy, multiscale entropy, and weighted entropy [37], [38], [39]. It is worth noting that the computation burden of these kinds of entropies may be heavy for the 4-D data, and the statistics for calculating entropy should be properly designed.

After the plugin of the spatiotemporal processing, the images are significantly enhanced, and the consequent processing of detection and recognition [11], [40] can be further applied. From this perspective, this spatiotemporal processing approach can be applied as a plug-and-play module to various radar imaging applications with time series.

The 4-D-imaging microwave radar with the joint spatiotemporal processing method greatly improves the SNCR of the radar images and makes it available for effectively detecting and localizing the survivors under the debris. Various results at CNTBUSR show that this processing makes the trapped victims appear with an SNR improvement of more than $20\,\mathrm{dB}$ compared with the original images overwhelmed by noise and clutter.

VI. Conclusion

A spatiotemporal processing method based on CF and EER has been proposed to enhance the images acquired by a 4-D imaging radar in the application of sensing trapped victims. The CF is applied in the spatial dimension of the 4-D radar image sequence and the EER in the temporal dimension. The joint processing method greatly improves the SNCR of the radar images and makes it available for effectively detecting and localizing the survivors through mounds of building debris. Real experiments in different conditions were carried out at CNTBUSR. The processed 4-D radar images with localization results and the extracted vital signs validated the performance of the proposed method. Moreover, the comparison results of different algorithms further verified the effectiveness of the spatiotemporal joint processing method. The experiments approximating the real postdisaster environment show the potential of the 4-D imaging radar together with the proposed spatiotemporal processing method to save lives in urban rescues.

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Zhi Li received the B.S. degree in electronic engineering and the M.S. degree in information and communication engineering from the National University of Defense Technology, Changsha, China, in 2010 and 2012, respectively, where he is currently pursuing the Ph.D. degree in information and communication engineering.

He is a Lecturer with the Army Engineering University of PLA, China. His research interests include biomedical signal processing, radar imaging, and machine learning.



Tian Jin (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in information and communication engineering from the National University of Defense Technology, Changsha, China, in 2002, 2003, and 2007, respectively.

From 2014 to 2015, he was a Visiting Scholar with Delft University of Technology, Delft, The Netherlands. He is a Professor with the National University of Defense Technology. He has authored or coauthored more than 100 peer-reviewed papers in international journals and conferences. His research interests include radar imaging, automatic target detection, and machine learning.

Dr. Jin was awarded the National Excellent Doctoral Dissertation of China in 2009 for his Ph.D. dissertation.

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Lianlin Li (Senior Member, IEEE) was born in 1980. He received the Ph.D. degree from the Institute of Electronics, Chinese Academy of Sciences, Beijing, China, in 2006.

He is currently a Hundred Talented Program Professor with Peking University, Beijing, China. His research interests include super-resolution imaging, microwave imaging, sparse signal processing, and ultrawideband radar systems.



Yongpeng Dai received the B.S. degree in electronic information engineering from the Harbin Institute of Technology, Harbin, China, in 2014, and the M.S. and Ph.D. degrees in information and communication engineering from the National University of Defense Technology, Changsha, China, in 2017 and 2021 respectively.

He is currently a Lecturer with the College of Electronic Science and Technology, National University of Defense Technology. His research interests include radar imaging, radar image improvement, automatic target detection, and neural network.



Yongping Song received the B.S. degree in electronic engineering and the M.S. and Ph.D. degrees in information and communication engineering from the National University of Defense Technology, Changsha, China, in 2012, 2014, and 2019, respectively.

He has been a Lecturer with Air Force Early Warning Academy, Wuhan, China, since 2021. He is currently an Assistant Researcher with the National University of Defense Technology. His research interests include MIMO radar image formation, radar target detection, and radar antijamming.



Yongkun Song was born in Henan, China, in 1993. He is pursuing the Ph.D. degree with the School of Electronic Science, National University of Defense Technology, Changsha, China.

His main research interests include radar signal processing, remote-sensing image processing, and deep learning.



Xiaolong Zhou (Student Member, IEEE) is pursuing the Ph.D. degree with the National University of Defense Technology, Changsha, China.

His research interests include radar signal processing and human pose estimation based on radar.

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